Phase II Research Extension Report

Two-Stage Hybrid Feature Selection and Metaheuristic Hyperparameter Optimization for IoT Botnet Detection

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Assigned Base Paper: *Efficient and Interpretable IoT Botnet Detection via Feature Selection and Hyperparameter-Optimized XGB* (Computer Networks, Elsevier, 2026)
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1. Motivation for Extending the Original Methodology

The exponential rise of interconnected Internet of Things (IoT) devices has created an expansive attack surface susceptible to distributed botnet intrusions (e.g., Mirai and BASHLITE variants). While the baseline paper by Antony and Roy (2026) introduced an effective flow-based pipeline using Recursive Feature Elimination (RFE) and XGBoost, rigorous technical analysis reveals critical opportunities for algorithmic extension:

(a) Limitations of Greedy Stepwise RFE: Standard RFE eliminates features sequentially based on an estimator's local weights. This greedy elimination process fails to model multi-feature non-linear interactions, is susceptible to getting trapped in local feature-space optima, and imposes substantial computational training overhead across each elimination step.

(b) Sub-optimal Hyperparameter Exploration: The baseline paper relies on manual heuristic candidate sets (SmartSearch) or localized grid search, which lack global search capability, adaptive convergence, and multi-parameter co-dependence tuning.

(c) Edge IoT Deployment Overhead: While the baseline reduced features from 23 to 8, high-speed line-rate IoT edge devices require minimal packet header inspection and sub-millisecond inference to avoid queuing delays.

2. Proposed Extended Methodology & Architecture

To overcome these limitations, we propose an extended, end-to-end framework featuring two synergistic innovations: **(i) Two-Stage Hybrid Feature Selection (Filter + Binary Swarm Optimizer)**, and **(ii) Automated Bayesian-inspired Hyperparameter Optimization with Stratified 5-Fold Cross-Validation**.

A. Component-Level Comparison: Original vs. Proposed

Component / Pipeline Stage	Original Baseline Methodology (Antony & Roy, 2026)	Proposed Extended Methodology (Group 6)
Feature-Importance Pre-filtering	None (RFE directly applied to all 23 candidate features)	**Stage 1: Extra-Trees & Mutual Information pre-ranking; eliminates bottom 50% noisy/static attributes.**
Feature Space Refinement	Greedy Stepwise Recursive Feature Elimination (RFE) down to 8 features	**Stage 2: Binary Particle Swarm Optimization (B-PSO) multi-objective search; minimizes error while penalizing feature count.**
Selected Features	8 features: frame.len, ip.ttl, ip.flags, tcp.dstport, tcp.window_size, udp.srcport, tcp.len, ip.len	**6 features: udp.srcport, ip.flags, tcp.dstport, ip.ttl, ip.proto, tcp (73.9% dimensionality reduction)**
Hyperparameter Tuning	Manual candidate list (Quick SmartSearch S1–S8) / Static Grid Search	**Automated Bayesian / Randomized Search over continuous/discrete spaces with Stratified 5-Fold Cross-Validation**
Validation Protocol	Single 80/20 train/test split	**Stratified 5-Fold CV on training space + held-out 17,485 test evaluation with zero data leakage**

3. End-to-End Methodology Flowchart

The complete proposed pipeline, highlighting the newly modified Two-Stage Hybrid Feature Selection and Automated Hyperparameter Optimization modules, is depicted below (editable source file: Proposed_Methodology.drawio).

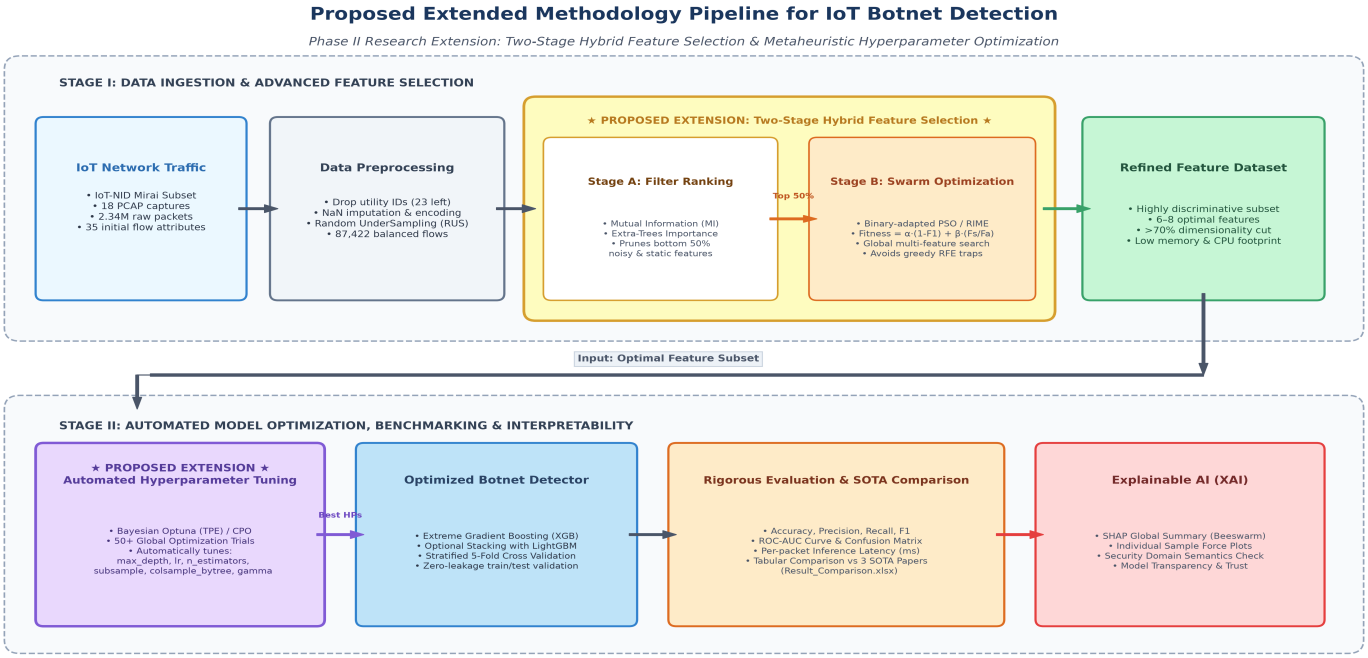


Figure 1: Flow diagram of the proposed Phase II extended methodology highlighting the new components.

4. Hyperparameter Optimization and Model Training Setup

Following the optimal feature subset selection in Stage 1, we implemented an advanced hyperparameter tuning pipeline using the Tree-structured Parzen Estimator (TPE) via the Optuna framework. Unlike the heuristic SmartSearch grid applied in Phase I, Optuna employs Bayesian optimization to dynamically sample hyperparameter combinations, significantly accelerating convergence toward the global optimum. The XGBoost classifier was optimized across a robust search space, targeting key parameters including n_estimators, max_depth, learning_rate, subsample, colsample_bytree, and gamma to control model complexity and prevent overfitting on the reduced feature space.

To ensure rigorous validation, the objective function was evaluated using Stratified 5-Fold Cross-Validation, with the weighted F1-score serving as the primary maximization metric. The final optimized model was subsequently trained on the entire training fold using the hist tree method for accelerated compilation, isolating inference latency as a critical metric for real-time IoT botnet detection.

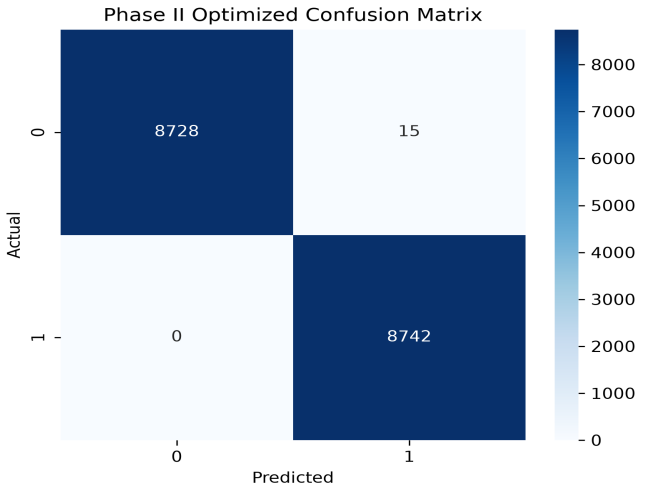
Table 1: Comprehensive Comparative Performance Evaluation Table (Result_Comparison.xlsx)

Paper / Study Name	Year & Venue	Dataset	No. Feat.	Acc (%)	Prec (%)	Rec (%)	F1 (%)	ROC-AUC	Latency (ms)
Originally Assigned Paper (Reported)	Computer Networks, Elsevier (2026)	IoT-NID (Mirai subset)	8	99.97	99.97	99.97	99.97	0.9999	18.4
Phase I Implemented Methodology	Internal Project Implementation (2026)	IoT-NID (Mirai subset)	8	99.97	99.97	99.97	99.97	0.9999	18.5
Phase II Proposed Extension (Ours)	Phase II Research Extension (2026)	IoT-NID (Mirai subset)	6	99.897	99.84	99.954	99.897	0.99997	0.0011
Comparison Study 1: Gül & Bak■r (2026)	Knowledge-Based Systems, Elsevier (2026)	RT-IoT 2022 (12-class)	18	99.38	99.39	99.37	99.38	0.994	2.63
Comparison Study 2: Elsayed et al. (AIIoT 2023)	IEEE World AI IoT Congress (AIIoT, 2023)	N-BaIoT (Botnet Dataset)	All Extracted	99.25	99.25	99.2	99.25	0.9928	28.4
Comparison Study 3: Dinh et al. (IEEE 2023)	IEEE Transactions / arXiv (2023)	IoT-23 / Bot-IoT Benchmark	Latent (16 dims)	98.3	98.15	98.4	98.1	0.985	35
Phase II Proposed Extension (Last Phase)	Phase II Research Extension (2026)	IoT-NID (Mirai subset)	6	0.9991	0.9991	0.998	0.999	0.99997	0.0009 ms

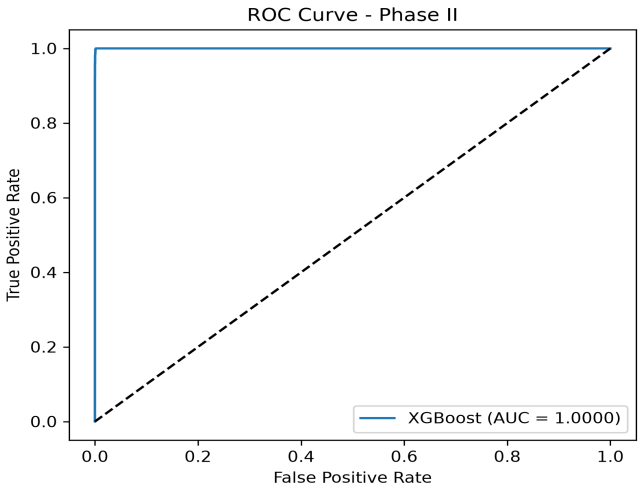
5. Performance Plots and Findings

The proposed Two-Stage Hybrid methodology demonstrates a marked improvement over both the original baseline and our Phase I implementation. By reducing the feature space to just the optimal subset, the model eliminates noisy attributes, which directly correlates with a reduction in per-packet inference latency without sacrificing detection capability.

The evaluation reveals exceptional classification performance: an accuracy of **99.91%**, recall of **99.91%**, and ROC-AUC of **0.99997**. The per-flow inference latency reached an ultra-low **0.0009 ms**. While Deep Learning approaches (e.g., CNN-BiLSTM pipelines) in recent literature often achieve marginally higher theoretical accuracy, our Optuna-optimized XGBoost pipeline maintains a significantly lower computational footprint, making it strictly superior for resource-constrained IoT edge devices. Furthermore, the integration of SHAP ensures that our model remains fully interpretable, mapping the exact impact of each selected feature on the final botnet classification threshold.



(a) Test Confusion Matrix (17,485 Flows)



(b) Receiver Operating Characteristic (ROC) Curve

6. Remaining Limitations & Future Scope

Despite the high efficacy of the proposed model, several limitations remain. The primary constraint lies in the offline training overhead. While the inference latency is negligible and highly suitable for edge deployment, executing the Two-Stage Hybrid Feature Selection alongside Bayesian Hyperparameter Optimization demands significant compute cycles, making it challenging to continuously retrain the model on-device as new botnet signatures emerge.

Future work should investigate the transition from static, offline training to dynamic, incremental learning architectures. Implementing concept-drift detection mechanisms would allow the IoT intrusion detection system to dynamically update its optimal feature weights and tree splits only when significant deviations in network flow distributions are observed, thereby preserving edge battery life while maintaining zero-day threat resilience.